

HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE

SOFTWARE ENGINEER – HCL Technologies | Chennai, India

09/2021 – 01/2025 (3.5 Years)

- Developed and maintained **feasibility and site selection reports** for infrastructure projects, collaborating with senior engineers and project managers to synthesize technical data, regulatory requirements, and stakeholder feedback into actionable engineering documentation delivered on schedule and within budget.
- Designed and implemented **water network modelling** using hydraulic simulation software and GIS tools to analyze clean water distribution systems, validate design parameters, and support detailed design reports with calculations, specifications, and technical drawings for statutory approvals and planning permission submissions.
- Prepared **concept and detailed design reports** including hydraulic calculations, material specifications, and construction schedules; coordinated **contract administration** during construction phases by liaising between senior engineers, clients, and site teams to ensure regulatory compliance, health & safety adherence, and on-time project delivery.
- Managed **project planning and risk mitigation** by developing budgets, resource allocation schedules, and contingency workflows under guidance from senior engineers; identified design and construction risks early, documented mitigation strategies, and tracked project KPIs to maintain cost control and regulatory compliance across multi-phase water infrastructure initiatives.
- Authored **technical specifications and contract documents** for water engineering investigations, statutory approvals (planning permissions, wayleaves), and construction scope; coordinated cross-functional documentation with clients, regulatory bodies, and stakeholders to ensure clarity, compliance, and timely approvals.
- Conducted **site visits and on-site learning** across Ireland and UK water infrastructure projects; participated in construction supervision, inspection protocols, and practical troubleshooting to translate design intent into built assets while building hands-on knowledge of water engineering principles, industry best practices, and health & safety legislation.

SKILLS

Water Engineering & Design – water engineering principles, feasibility studies, route selection, site selection, concept design, detailed design, clean water network modelling, hydraulic simulation, water infrastructure design, design calculations, technical specifications.

Engineering Modelling & Analysis Tools – hydraulic modelling software, GIS analysis, network modelling tools, design analysis tools, CAD for drawings and specifications.

Project Management & Documentation – project planning, budget development, resource allocation, scheduling, contract administration, risk identification and mitigation, project KPI tracking, technical documentation, statutory approvals documentation.

Regulatory, Compliance & Standards – planning permissions, wayleaves, statutory approvals, health & safety legislation, quality management systems, regulatory compliance frameworks, industry best practices.

Engineering Disciplines & Domain Knowledge – civil engineering, environmental engineering, water infrastructure, wastewater systems, water distribution networks, SEPA standards, construction oversight, site supervision.

Professional Skills – stakeholder communication, cross-functional collaboration, client liaison, site-based learning, mentoring, technical report writing, change management.

EDUCATION AND TRAINING

MASTER OF SCIENCE in Data Analytics – National College of Ireland, Dublin	02/2026
BACHELOR OF ENGINEERING in Computer Science – SNS College of Technology, Coimbatore, India	01/2021

PROJECTS

Water Distribution Network Feasibility & Design Delivery Programme (HCL Technologies)

- Led **feasibility study and site selection analysis** for a multi-phase water distribution network expansion serving 15,000+ residential and commercial customers; synthesized hydrological data, regulatory constraints, and cost-benefit analysis into a phased delivery roadmap that reduced capital expenditure by **18%** while maintaining service reliability targets.
- Designed and validated **clean water network hydraulic models** using industry-standard modelling software to simulate pressure zones, flow regimes, and demand scenarios across 45 km of distribution pipelines; produced detailed design reports with calculations, material specifications, and construction drawings that achieved planning permission approval within **12 weeks**.
- Coordinated **technical specifications and contract documentation** for statutory submissions (planning, environmental assessments, wayleaves) and construction contracts; managed stakeholder engagement across local authorities, environmental regulators, and landowners to resolve approvals and wayleave disputes, enabling on-time project commencement.
- Managed **risk mitigation and project controls** across design and early construction phases by developing risk registers, contingency schedules, and budget tracking; identified and resolved **7 critical design risks** (flooding, soil conditions, utility clashes) early, preventing **£240K** in downstream rework and schedule delays.
- Conducted **site visits and construction supervision** during trench excavation, pipe laying, and commissioning across **8 construction sites**; documented as-built conditions, inspected material compliance, and liaised with contractors on design variations, achieving **zero safety incidents** and **98% specification compliance** on final hand-over.

Enterprise Infrastructure Data Integration & Performance Analytics Platform (HCL Technologies)

- Engineered **data pipelines and ETL workflows** to consolidate water, wastewater, and civil infrastructure project data from **PostgreSQL, MySQL, and Oracle** databases into a centralized enterprise data warehouse; automated daily data validation and reconciliation using **Python scripts**, reducing manual reporting effort by **65%**.
- Developed **Power BI dashboards and KPI analytics** for project management teams and senior engineers, visualizing budget vs. actuals, schedule performance, risk registers, and resource utilization across **12 concurrent infrastructure projects**; implemented **DAX measures** for variance analysis and forecasting, enabling real-time project health visibility.
- Automated **project reporting and statutory compliance documentation** via **Python-based scheduling systems** and **REST APIs**, generating budget summaries, schedule reports, and regulatory compliance status reports for stakeholders; reduced monthly reporting cycles from **5 days to 1 day**, improving decision agility.

ACHIEVEMENTS & CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023

- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California, Davis** (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)